

Week 15 Pre-Lab Quiz — Capstone Readiness

CE 310 | 10 questions | 10 points | Due before lab session

This quiz reviews the statistical methods you will apply in the Week 15 capstone lab. You will need regression, hypothesis testing, sensitivity, and Monte Carlo skills from Weeks 9–13.

Q1. A multiple regression model gives $\text{Cooling_kWh} \sim 0.82 \cdot \text{Outdoor_Temp_F} + 3.45 \cdot \text{BH}$. The BH coefficient is +3.45. What does this mean?

- A. For every 1°F increase in outdoor temperature, cooling increases by 3.45 kWh
- B. Business hours cause cooling to increase by 3.45 kWh relative to the same outdoor temperature outside business hours
- C. Removing business hours reduces annual cooling by 3.45 kWh
- D. The model predicts 3.45 kWh of cooling when both Outdoor_Temp_F and BH equal zero

Q2. Model 1 (simple regression on temperature) has $R^2 = 0.72$. Model 2 (adds a BH indicator) has $R^2 = 0.73$. Which conclusion is best supported?

- A. Model 2 is always better because a higher R^2 is strictly better
- B. The BH variable adds very little explanatory power beyond temperature alone
- C. Model 1 should be preferred because adding variables always overfits
- D. R^2 cannot be used to compare models with different numbers of predictors

Q3. A multiple regression model for a small hotel returns a BH coefficient of -3.27 . The hotel's standard business hours are M-F 9 AM–5 PM. Which interpretation is most physically plausible?

- A. The model is incorrect and should be re-estimated
- B. At the same outdoor temperature, daytime M-F hours use less cooling than evening and weekend hours, because hotel guest demand peaks in evenings
- C. Hotels have no cooling demand during business hours
- D. The negative sign means the BH variable was entered in the wrong direction in the formula

Q4. A two-sample t-test compares mean cooling during business hours (mean = 34.0 kWh/hr) vs. non-business hours (mean = 29.8 kWh/hr) and gives $p = 8.45 \times 10^{-42}$. The regression BH coefficient is -3.27 . Why do the t-test and regression disagree on the direction of the BH effect?

- A. The t-test uses a different dataset than the regression
 - B. The t-test does not control for outdoor temperature; the regression does
 - C. The p-value is too small for the t-test to be valid
 - D. The regression formula should use OLS; the t-test uses a different estimator
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Q5. In a Monte Carlo cost simulation, you sample electricity rate from $N(0.12, 0.01)$ and temperature uncertainty from $N(0, 2^\circ\text{F})$. The resulting annual cost distribution has $P10 = \$28,700$ and $P90 = \$36,800$. What does this range represent?

- A. The cost will definitely fall between \$28,700 and \$36,800
 - B. The 80% confidence interval for annual cost given input uncertainty
 - C. The range of costs observed in the historical data
 - D. The minimum and maximum possible annual costs
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Q6. You run an OAT (one-at-a-time) sensitivity analysis on a regression model and find that a ± 1 standard deviation change in outdoor temperature changes annual cost by $\pm \$8,200$, while a ± 1 standard deviation change in the electricity rate changes annual cost by $\pm \$7,100$. What does this suggest?

- A. Temperature is the dominant driver of cost uncertainty
 - B. Electricity rate is the dominant driver of cost uncertainty
 - C. Both inputs contribute approximately equally to cost uncertainty
 - D. Neither input drives cost uncertainty — the model intercept dominates
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Q7. You add a second predictor to your regression model. The R^2 increases from 0.811 to 0.813, but the adjusted R^2 decreases from 0.811 to 0.810. What should you conclude?

- A. The new predictor is statistically significant and should be kept
 - B. The new predictor adds almost no explanatory power after penalizing for model complexity
 - C. The model is overfit and should be rebuilt with fewer observations
 - D. Adjusted R^2 is not a valid metric for models with more than two predictors
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Q8. A warehouse has annual cooling of 149,000 kWh and an electricity rate of \$0.12/kWh. You simulate 10,000 scenarios with rate $\sim N(0.12, 0.01)$ and annual energy $\sim N(149000, \sigma_{\text{kWh}})$. Which of the following correctly computes the annual cost for each Monte Carlo sample?

- A. $\text{cost} = 149000 * 0.12$
- B. $\text{cost} = \text{kWh_samples} * \text{rate_samples}$
- C. $\text{cost} = \text{kWh_samples} * 0.12$
- D. $\text{cost} = 149000 * \text{rate_samples} + \text{kWh_samples}$

Q9. Two capstone teams both use temperature as a predictor variable, but analyze completely different topics (one studies cooling energy, another studies traffic volume). Both relationships are statistically significant, but the two teams get very different R^2 values. What best explains this?

- A. One of the two teams made a calculation error
- B. Different domains have different amounts of unexplained variance — the same predictor can be a strong driver in one context and a weak one in another, even with identical regression methodology
- C. R^2 should always come out identical if the regression is done correctly
- D. Random noise in data collection accounts for all of the difference

Q10. After fitting a regression model, you plot residuals (actual – predicted) vs. fitted values and see a fan shape (residuals spread wider at high fitted values). What does this indicate?

- A. The model has high R^2 and is a good fit
 - B. The model may have heteroskedasticity — variance of errors increases with the fitted value
 - C. The model has too many predictors
 - D. The model is correctly specified and the fan shape is expected
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